

HARMONICS PITCH Technical Manual

Harmonic structure, pitch, and instrument generation

Shape harmonic emphasis, rhythmic spectral motion, or generated instrument voices.

Included products: THOTH, TEHUTI, AU

Routing authority: Thoth and Tehuti are insert safe. AU is the family instrument and synth.

Canon baseline

Normalized control 1 equals 0.809016994 ($\text{PHI}/2$). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

THOTH

Sacred harmonic EQ with low and high shelves, visible shelf curves, Flower of Life mode, protected I/O, and parallel mix.

Declared product: Thoth | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
input	Input	Float
mix	Mix	Float
ls_gain	Low Shelf	Float
hs_gain	High Shelf	Float
output	Output	Float
flower	Flower of Life	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

source/Thoth.png

Meter and visualizer semantics

Display	What it measures	How to read it
Spectrum bands	Live energy in each displayed EQ region	Use to see where harmonic energy is accumulating.
Shelf curves	Current low/high shelf response	These show the selected EQ shape, not measured level.
IN / OUT	Pre/post peak activity	Use for level-fair harmonic decisions.

TEHUTI

Harmonic motion processor using low, mid, and high ducking with depth, speed, mix, and Flower of Life mode.

Declared product: \${PROJECT_NAME} | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
low_duck	Low Duck	Float
mid_duck	Mid Duck	Float
high_duck	High Duck	Float
depth	Depth	Float
speed	Speed	Float
mix	Mix	Float
flower	Flower of Life	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

source/Tehuti.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Main-program peak activity	Shows the signal being ducked.
SIDECHAIN SEND	External sidechain envelope	Confirms that the host sidechain is present.
LOW GR	Reduction below 162 Hz	Shows low-band ducking driven by the sidechain.
MID GR	Reduction from 162 Hz to 1.62 kHz	Shows mid-band ducking driven by the sidechain.
HIGH GR	Reduction above 1.62 kHz	Shows high-band ducking driven by the sidechain.
Duck curves	Expected reduction versus sidechain energy	The live dot shows the current operating point.
History	Recent sidechain and reduction motion	Use to judge pumping and release behavior.

AU

DIHWorld instrument and synth with factory sound selection, drag-and-drop source support, nine Neter voices, modulation, space, glide, sustain, and master control. The current AU is registered and uses smoothed output normalization.

Declared product: AU | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Instrument Track | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
SOUND	Factory Sound	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
PTAH	PTAH	Referenced
RA	RA	Referenced
SET	SET	Referenced
ASAR	ASAR	Referenced
MAAT	MAAT	Referenced
SEKHMET	SEKHMET	Referenced
KHNUM	KHNUM	Referenced
HERU	HERU	Referenced
DRIVE	DRIVE	Referenced
AMOUNT	AMOUNT	Referenced
DEPTH	DEPTH	Referenced
SV_MIX	SV_MIX	Referenced
HARP	HARP	Referenced
ORGAN	ORGAN	Referenced
CHOIR	CHOIR	Referenced
MOD	MOD	Referenced
BREATH	BREATH	Referenced
VIBRATO	VIBRATO	Referenced
PHI	PHI	Referenced
AZIMUTH	AZIMUTH	Referenced
FLOWER	FLOWER	Referenced
HATHOR	HATHOR	Referenced
RAW	RAW	Referenced
MOD_MACRO	MOD_MACRO	Referenced
MASTER	MASTER	Referenced
GLIDE	GLIDE	Referenced
PURE	PURE	Referenced
SUSTAIN	SUSTAIN	Referenced
GAINMATCH	GAINMATCH	Referenced

Artwork authority

Assets/AU_Background.png

OutputNormalizer is prepared before processing and advances rampStep per sample, preventing block-stepped gain changes. The installed Audio Unit passes registration/validation.